Name - Krishna Lokhande

Roll No. 32

PRN - 12520011

Course - Computer Networks

Assignment 4

Write a program for error detection and correction for 7/8 bits ASCII codes using Hamming codes or CRC

1. Aim

To write and execute a C program for error detection and correction using Hamming codes for 7/8-bit ASCII data, and to generate the corresponding Hamming code by calculating parity bits.

2. Theory

Hamming code is an error-detecting and error-correcting technique used in digital communication and computer networks. It adds parity bits to data bits so that a single-bit error can be detected and corrected at the receiver.

For 4 data bits, the Hamming (7,4) code uses 7 total bits: 4 data bits and 3 parity bits. The parity bits are placed at positions 1, 2, and 4.

Bit arrangement:

1	2	3	4	5	6	7
L1	L2	D2	L4	D1	D3	D4

The parity bits are calculated using XOR operations:

- L1 = A[2] XOR A[4] XOR A[6]
- L2 = A[2] XOR A[5] XOR A[6]
- L4 = A[4] XOR A[5] XOR A[6]

Here, XOR gives 0 when the number of 1s is even and 1 when the number of 1s is odd. The resulting 7-bit sequence forms the Hamming code.

3. Program

```
#include <stdio.h>

int main() {
   int A[7], L1, L2, L4;

   printf("Enter 4 data bits: ");

   scanf("%d", &A[2]);
   scanf("%d", &A[4]);
```

```
    scanf("%d", &A[5]);
    scanf("%d", &A[6]);

    L1 = A[2] ^ A[4] ^ A[6];
    L2 = A[2] ^ A[5] ^ A[6];
    L4 = A[4] ^ A[5] ^ A[6];

    A[0] = L1;
    A[1] = L2;
    A[3] = L4;

    printf("\nHamming Code => ");
    int i;
    for (i = 0; i < 7; i++) {
        printf("%d", A[i]);
    }

    return 0;
}
```

4. Test Cases and Output

The following test cases were executed using the above program. The output screenshots are included as evidence of execution.

Test Case	Input Data Bits	Expected Hamming Code	Observed Output
1	0 1 1 0	1100110	1100110
2	0 0 0 1	1101001	1101001
3	0 0 1 1	1000011	1000011

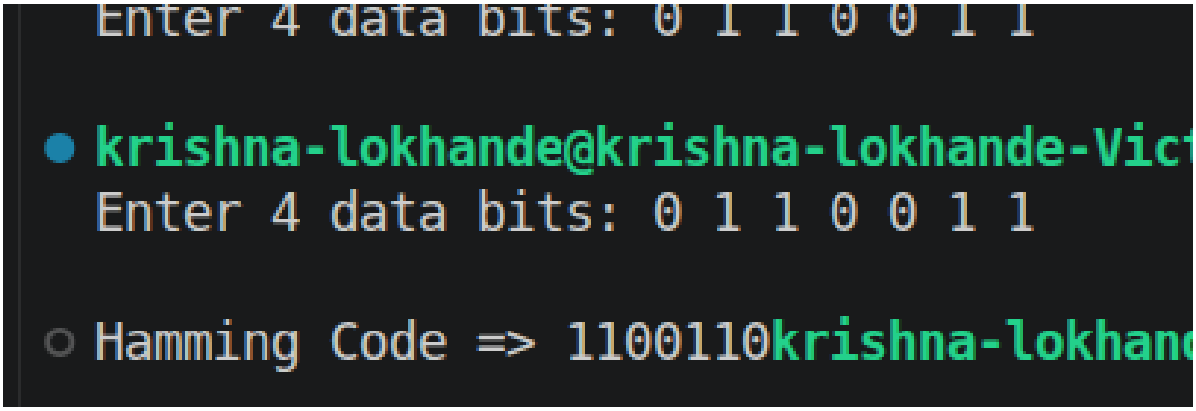


Figure 1: Test Case 1 Output

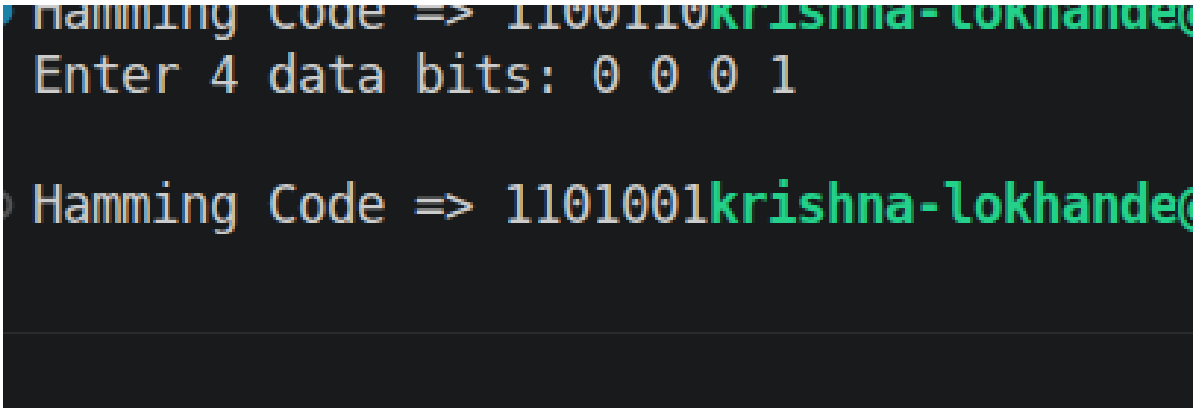


Figure 2: Test Case 2 Output

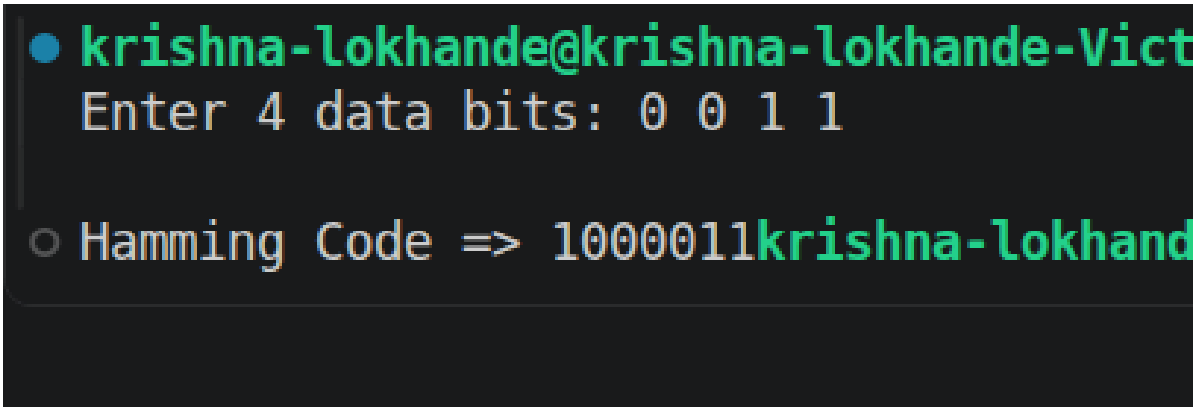


Figure 3: Test Case 3 Output

5. Result

The C program successfully generated 7-bit Hamming codes from the given 4-bit data inputs. The calculated parity bits were inserted at the required positions, producing the expected Hamming code for each test case.

6. Conclusion

Thus, the Hamming (7,4) coding technique was successfully implemented in C. The program demonstrates how parity bits can be generated using XOR operations and combined with data bits to form a Hamming code. Hamming coding is useful for detecting and correcting single-bit errors during data transmission.